

Signed Prefixes and the Distinguished Zero Mode

Sharp Abel-BV Duality, an Exact Exponent Transfer, and the Remaining Fixed-Shift Input

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Abstract

TPC-111 showed that the distinguished determinant zero mode survives every provenance-preserving coarsening and proved the sharp signed-prefix/BV duality. We turn that finite criterion into a complete exponent ledger. For a growing family, a literal prefix saving $X^{-\delta}$, an outer BV-envelope cost X^{ℓ_Z} , and a content-remainder exponent η_{cont} imply

$$\eta_Z^{\text{cert}} = \min\{\delta - \ell_Z, \eta_{\text{cont}}\}.$$

Here $\delta \geq \ell_Z$ is required; otherwise this certificate provides no nonnegative zero-mode exponent for the TPC-112 cone. Combined with the TPC-112 determinant compatibility cone, sufficient thresholds for this Abel/triangle certificate are

$$\delta \geq \ell_Z + \frac{\lambda_D}{2}, \quad \eta_{\text{cont}} \geq \frac{\lambda_D}{2}.$$

We also give the minimum rule for a subpower family of differently scaled fiber classes. This is an exact transfer statement, not a proof of any growing input. The finite duality is imported and restated self-contained; the typed fixed- h_0 ledger is L0/L1. No L2 cancellation theorem is claimed.

1 The literal ordered zero-mode pairing

Fix the prescribed nonzero shift h_0 . At the verified outer-reassembly interface of TPC-111—that is, conditional on its content, masks, polarizations and native keys being retained and the literal outer reassembly being verified—the literal zero mode is indexed by outer affine keys $\theta \in \Theta_X$, with canonically ordered intervals

$$\mathcal{I}_{\theta,X} = \{r_{\theta,1} < \cdots < r_{\theta,m_\theta}\},$$

and writes

$$Z_X = \sum_{\theta \in \Theta_X} c_{\theta,X} \sum_{i=1}^{m_\theta} W_{\theta,X}(r_{\theta,i}) \sigma_{\theta,X}(r_{\theta,i}) + E_{\text{content},X}^{(111)}$$

[3]. Choose a predeclared retained set $\mathcal{F}_X \subseteq \Theta_X$. For $f = \theta \in \mathcal{F}_X$, set

$$t_{f,i} = r_{\theta,i}, \quad \sigma_{f,i} = \sigma_{\theta,X}(r_{\theta,i}), \quad w_{f,i} = c_{\theta,X} W_{\theta,X}(r_{\theta,i}).$$

Define $E_{\text{cont},X}$ to be the original content term plus the exact contributions of every outer key in $\Theta_X \setminus \mathcal{F}_X$. Then the complete literal zero mode is

$$Z_X = \sum_{f \in \mathcal{F}_X} Z_f + E_{\text{cont},X}, \quad Z_f = \sum_{i=1}^{m_f} \sigma_{f,i} w_{f,i}.$$

Thus $\sigma_{f,i}$ retains the literal arithmetic sign and affine data, while $w_{f,i}$ retains the outer coefficient and reconstruction multiplier. The retained/content partition and this factor allocation are fixed before applying the estimate. Moving a rough arithmetic factor between the two changes the gate and must be recorded.

Define signed prefixes

$$S_{f,k} = \sum_{i=1}^k \sigma_{f,i}, \quad \Delta_f = \max_{1 \leq k \leq m_f} |S_{f,k}|,$$

and the endpoint-anchored variation norm

$$\|w_f\|_{\text{BV}^*} = |w_{f,m_f}| + \sum_{k=1}^{m_f-1} |w_{f,k} - w_{f,k+1}|.$$

2 Imported Abel identity and sharp duality

The next theorem is the finite duality already proved in TPC-111. We include its short proof to fix the exact norm used by the exponent ledger; it is not presented as a second new duality theorem.

Lemma 2.1 (Finite Abel identity). *For every complex pair of sequences $(\sigma_i)_{i=1}^m$ and $(w_i)_{i=1}^m$,*

$$\boxed{\sum_{i=1}^m \sigma_i w_i = S_m w_m + \sum_{k=1}^{m-1} S_k (w_k - w_{k+1})}, \quad S_k = \sum_{i=1}^k \sigma_i.$$

Proof. Use $\sigma_i = S_i - S_{i-1}$, with $S_0 = 0$, and telescope. □

Theorem 2.2 (Sharp signed-prefix/BV duality). *For a fixed signed sequence $\sigma = (\sigma_i)_{i=1}^m$,*

$$\boxed{\sup_{\|w\|_{\text{BV}^*} \leq 1} \left| \sum_{i=1}^m \sigma_i w_i \right| = \max_{1 \leq k \leq m} |S_k|}.$$

Proof. [Lemma 2.1](#) and the triangle inequality give the upper bound. Choose k_* maximizing $|S_k|$. If $S_{k_*} \neq 0$, set

$$\omega = \frac{\overline{S_{k_*}}}{|S_{k_*}|}, \quad w_i = \begin{cases} \omega, & i \leq k_*, \\ 0, & i > k_*. \end{cases}$$

If $k_* < m$, the norm has one jump of size one and zero endpoint; if $k_* = m$, it has endpoint one and no jump. Thus $\|w\|_{\text{BV}^*} = 1$, while the pairing equals $\omega S_{k_*} = |S_{k_*}|$. The zero sequence is immediate. □

Remark 2.3 (Fiberwise sharpness). The theorem is sharp for each fixed fiber. After summing fibers, the triangle inequality

$$|Z_X| \leq \sum_f \Delta_f \|w_f\|_{\text{BV}^*}$$

is a sufficient outer certificate, not a necessary one: additional inter-fiber cancellation may occur.

3 The growing exponent transfer

Let $A_f \geq 0$ be a literal comparison scale for the signed coefficient on fiber f . The future prefix theorem and the outer reassembly theorem must be stated separately.

Definition 3.1 (Typed prefix and envelope inputs). For exponents $\delta, \ell_Z \geq 0$, define:

$$\text{(Prefix)} \quad \Delta_f \leq X^{-\delta+o(1)} A_f \quad \text{uniformly on the retained actual outer fibers;} \quad (1)$$

$$\text{(Envelope)} \quad \sum_{f \in \mathcal{F}_X} A_f \|w_f\|_{\text{BV}^*} \leq X^{\ell_Z+o(1)} Q_X^2. \quad (2)$$

In addition, a complete certificate must record

$$|E_{\text{cont},X}| \leq X^{-\eta_{\text{cont}}+o(1)} Q_X^2 \quad (3)$$

for some $\eta_{\text{cont}} \geq 0$. All three statements use the same factor allocation and quantifier range.

Theorem 3.2 (Prefix-to-zero-mode transfer). *If (1)–(3) hold and $\delta \geq \ell_Z$, then*

$$\boxed{|Z_X| \leq X^{-\eta_Z^{\text{cert}}+o(1)} Q_X^2,}$$

where the admissible zero-mode exponent is

$$\boxed{\eta_Z^{\text{cert}} = \min\{\delta - \ell_Z, \eta_{\text{cont}}\}.}$$

Proof. Apply [theorem 2.2](#) fiberwise and sum:

$$|Z_X| \leq \sum_f \Delta_f \|w_f\|_{\text{BV}^*} \leq X^{-\delta+o(1)} \sum_f A_f \|w_f\|_{\text{BV}^*}.$$

Now use the envelope input and add the content remainder. The slower of the two powers controls the sum. $\square$

Remark 3.3 (A negative displayed exponent). If $\delta < \ell_Z$, the prefix branch still gives $X^{-(\delta-\ell_Z)+o(1)} Q_X^2$, but it does not certify a nonnegative saving exponent. One must not enter a negative η_Z^{cert} into the TPC-112 cone.

4 Compatibility with determinant energy

TPC-112 requires

$$D_X \geq X^{-\lambda_D+o(1)} Q_X^3 J_X^2, \quad |Z_X| \leq X^{-\eta_Z+o(1)} Q_X^2, \quad \lambda_D \leq 2\eta_Z$$

on the same literal post-bin coefficient [\[4\]](#).

Corollary 4.1 (Exact threshold for this Abel/triangle certificate). *Under the typed inputs of [definition 3.1](#), a separately certified determinant exponent λ_D is compatible whenever both*

$$\boxed{\delta \geq \ell_Z + \frac{\lambda_D}{2}, \quad \eta_{\text{cont}} \geq \frac{\lambda_D}{2}.}$$

The reserve in the determinant-zero cone is

$$\rho_{\text{det}} = 2 \min\{\delta - \ell_Z, \eta_{\text{cont}}\} - \lambda_D.$$

Proof. Insert the admissible exponent η_Z^{cert} from [theorem 3.2](#) into $\lambda_D \leq 2\eta_Z$. $\square$

Combining with the phase certificate of TPC-121, where one admissible recorded exponent is

$$\lambda_D^{\text{cert}} = (\lambda_E + 2\lambda_\phi - \gamma_R)_+,$$

gives the explicit joint target

$$\boxed{\begin{aligned} \delta &\geq \ell_Z + \frac{(\lambda_E + 2\lambda_\phi - \gamma_R)_+}{2}, \\ \eta_{\text{cont}} &\geq \frac{(\lambda_E + 2\lambda_\phi - \gamma_R)_+}{2}. \end{aligned}}$$

This equation is bookkeeping, not a joint arithmetic theorem [5].

Corollary 4.2 (Uniform subpower-class ledger). *Let $\{\mathcal{F}_{X,j} : j \in \mathcal{J}_X\}$ be a predeclared partition of the retained actual fibers. Suppose there are a common constant C , a common sequence $\varepsilon_X \rightarrow 0$, and nonnegative numbers $\delta_{X,j}, \ell_{X,j}$ such that*

$$|\mathcal{J}_X| \leq X^{\varepsilon_X},$$

$$\Delta_f \leq CX^{-\delta_{X,j} + \varepsilon_X} A_f \quad (f \in \mathcal{F}_{X,j}), \quad \sum_{f \in \mathcal{F}_{X,j}} A_f \|w_f\|_{\text{BV}^*} \leq CX^{\ell_{X,j} + \varepsilon_X} Q_X^2$$

for every $j \in \mathcal{J}_X$. Assume also the content estimate

$$|E_{\text{cont},X}| \leq CX^{-\eta_{\text{cont}} + \varepsilon_X} Q_X^2.$$

For any fixed $\eta_{\text{cls}} \geq 0$ satisfying

$$\eta_{\text{cls}} \leq \liminf_{X \rightarrow \infty} \inf_{j \in \mathcal{J}_X} (\delta_{X,j} - \ell_{X,j}),$$

where the infimum of the empty set is $+\infty$ (in that case the retained sum is zero), one admissible recorded exponent is

$$\boxed{\eta_Z^{\text{cert}} = \min\{\eta_{\text{cont}}, \eta_{\text{cls}}\}.$$

Proof. On class j , [theorem 2.2](#) and the two uniform bounds give

$$\sum_{f \in \mathcal{F}_{X,j}} |Z_f| \leq C^2 X^{-(\delta_{X,j} - \ell_{X,j}) + 2\varepsilon_X} Q_X^2.$$

The liminf condition makes $\delta_{X,j} - \ell_{X,j} \geq \eta_{\text{cls}} - o(1)$ uniformly in j . Summing at most X^{ε_X} classes and adding the content remainder proves the claim. $\square$

Remark 4.3 (Typed losses and no double charging). The quantity $\delta - \ell_Z$ and the content exponent η_{cont} have zero-mode/soft-remainder roles; λ_D has the determinant-energy role; and Λ_{phys} has the physical-reassembly role. A source identifier may not be spent in two roles unless a theorem supplies the corresponding joint inequality and records the dependency replacement. In particular, determinant reserve is not physical endpoint slack.

5 Why all prefixes are necessary

Proposition 5.1 (The final sum is insufficient). *For every $N \geq 1$ there is a real sequence of length $2N$ with*

$$S_{2N} = 0, \quad \max_{k \leq 2N} |S_k| = N.$$

Hence cancellation of the complete unweighted fiber sum does not control the zero-mode pairing against varying multipliers.

Proof. Take N entries equal to 1, followed by N entries equal to -1 . $\square$

Proposition 5.2 (Ordering is part of the certificate). *The multiset of signed atoms determines the final sum but not the maximum prefix. Therefore a prefix theorem must use the canonical literal order, or prove a permutation-uniform estimate strong enough to include it. Choosing a favorable order after seeing the signs is not a provenance-preserving argument.*

Proof. For N signs $+1$ and N signs -1 , grouping equal signs gives maximum prefix N , whereas alternating signs gives maximum prefix 1. $\square$

The signed sequence imported from TPC-107 and TPC-108 contains actual Möbius/Liouville-type factors [1, 2]. A bound for independent random signs, an averaged shift, or a rearranged sequence does not prove (1).

6 Audit certificate and next decision

The deterministic script `experiments/tpc122_prefix_transfer_audit.py` checks the complex Abel identity, the BV bound, the sharp step extremizer, the large-prefix/final-zero counterexample, the TPC-111 factor-allocation crosswalk, the content-aware compatibility/reserve ledger, and the uniform subpower-class minimum rule.

Statement	Level and status
Abel identity and sharp BV duality	Exact finite theorem (L0).
Crosswalk $f = \theta$, $t_{f,i} = r_{\theta,i}$ to the TPC-111 ordered affine outer fibers, with the declared factor allocation	Fixed- h_0 interface (L1).
Growing prefix input (1)	Not proved; parity-facing arithmetic input.
Outer BV envelope (2)	Not proved on the complete actual family.
Content remainder (3)	Not proved at the required exponent.
Positive zero-mode exponent or fixed- h_0 saving	Not proved; no L2 advance.

TPC-122 reduces the zero-mode side to three typed estimates and computes the exact saving delivered by this certificate. It does not establish any of them. A next paper should inspect the literal ordering and factor allocation on one actual fiber class; if the multiplier has polynomial BV cost, the signed prefix saving must exceed that cost before the determinant exponent is even considered.

Explicit exclusions. Nothing here proves H3, the actual prefix estimate, a Möbius or Liouville cancellation theorem, a parity breach, a prime-pair lower bound, a Hardy–Littlewood asymptotic, or the twin-prime conjecture. The strict physical endpoint $\Lambda_{\text{phys}} < 1/400$ remains independent.

References

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